Sandra Samuel, MBA, MSc

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Education

Rotman School of Management, University of Toronto, Toronto, ON — Sep 2011 – Dec 2013
Master of Business Administration (MBA)
- Concentration: Finance
- Dean's Scholarship (top 10% of cohort)
- Capstone project: Valuation of a mid-cap Canadian REIT
- Co-author, Rotman Finance Review, Issue 14

McGill University, Faculty of Science, Montreal, QC — Aug 2006 – Apr 2010
Bachelor of Science (Honours), Mathematics and Economics
- Minor: Computer Science
- Graduated with Distinction, GPA 3.8/4.0
- Recipient, Faculty of Science Undergraduate Research Award (2009)
- Thesis: Stochastic volatility models for short-rate dynamics

Professional Experience

Canadian Imperial Bank of Commerce, Toronto, ON — Jan 2014 – Mar 2021
Senior Analyst, Quantitative Strategies & Research
- Built and maintained the bank's proprietary yield curve and credit spread engine in Python and C++, used by 40+ trading desks across CIBC Capital Markets
- Developed a Monte Carlo framework for pricing callable structured notes, reducing overnight batch runtime from 6.5 hours to 1.2 hours
- Mentored 3 junior analysts and rotated them through three quarterly projects
- Presented model governance updates to the firm's Model Risk Committee twice per year

Bank of Montreal, Toronto, ON — Summer 2012, Summer 2013
Summer Analyst, Fixed Income Research
- Built a Nelson-Siegel-Svensson yield curve fitter in MATLAB for the government bond desk
- Wrote weekly commentary on Canadian sovereign and provincial spreads, circulated to 120 readers
- Presented two recommendations that were adopted into the desk's hedging framework

Technical Skills

Languages: Python (NumPy, SciPy, pandas, scikit-learn), C++, MATLAB, R, SQL, VBA, Excel (advanced)
Tools: Bloomberg Terminal, Git, Linux, AWS (EC2, S3), Docker, Jupyter, LaTeX
Domain: Fixed income, yield curve modeling, credit spreads, Monte Carlo methods, stochastic calculus, time series, Black-Scholes, Hull-White, Vasicek, CIR

Certifications

CFA Institute — Level I, Level II, Level III
- Level I: Jun 2012
- Level II: Dec 2012
- Level III: Jun 2013

Teaching Experience

Rotman School of Management, Toronto, ON — Sep 2016 – Apr 2019
Sessional Lecturer, Quantitative Methods for Finance
- Taught 2 sections of 60 students each, rating 4.6/5 average
- Redesigned the assignment bank from scratch, cut grading time by roughly 30

McGill University, Faculty of Science, Montreal, QC — Sep – Dec 2010
Teaching Assistant, Probability Theory
- Graded 40 hours per week for a 200-level probability course

Volunteer

Toronto Chapter, CFA Society Toronto — Jan 2020 – Present
Volunteer Mentor
- Mentored 4 CFA candidates through Level I preparation, 3 of them passed on first attempt

Publications

Samuel, S. (2013). "Stochastic volatility in the short-rate dynamics: evidence from Canadian government bond futures." *Rotman Finance Review*, Issue 14, pp. 41–63.
- Co-author: Prof. Michael Brennan, Rotman School of Management
- Cited in 2 subsequent working papers

Interests

Long-distance running (2:41 marathon, Toronto Waterfront, 2019), chess (FIDE rating 1820), reading economic history, hiking in the Canadian Shield, learning Portuguese (B1).